



Netflix Originals Delivery Specifications

version OC-3-3

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1 Objective

The Netflix streaming service offers an outstanding, high-quality viewing experience that preserves original artistic intent. This level of quality requires Netflix to work with source files that match the original frame rate and aspect ratio as it was originally shot. If Partner does not have the source Netflix requests, Netflix will work closely with Partner to see if there are acceptable alternatives. Netflix may be referred to as “Netflix” or “our” and Partner may be referred to as “you” and “your” in this schedule.

For each Title, all Source Material elements set forth in this schedule shall work in conjunction (conform) with one another as a properly functioning single asset. Source Material must be compliant with the specifications below detailed in this document.

This document lists specifications for deliverables to Netflix. However, some of them may not apply to your production. Please refer to your Netflix Delivery Requirements List or your Netflix Representative to verify which sections apply to your delivery.

2 Source Material Requirements

There are no exceptions to the following requirements, unless previously approved by your Netflix Representative.

- All video must consist of the feature program with one (1) second of black and silence at the head and tail of program.
- No commercial blacks can be contained in the program.
- Deliveries must not contain bars and tones, bumpers, segment recaps or previews, advertisements, slates, ratings cards, FBI warning cards, VITC time code, placards, overlay branding or website link callouts before, during and/or after the program. URLs present during end credits are permissible.
- Our requirement is to receive a **non-subtitled version** of the content. Netflix defines “non-subtitled” as the presence of main titles, end credits, narrative text, location call-outs, and other supportive/creative text, but **no burned-in subtitled dialog, regardless of the language** in the primary video.
- Source material must be **uncensored**.
- Translations within secondary audio and timed text assets should not introduce slurs that were not present in the original language or unnecessarily harshen/soften the language used, unless specifically requested by Netflix.
- Netflix only accepts **IMF Application 2 Extended** files for primary AV delivery.
- The IMF package must be created from the Video Display Master (VDM), or directly out of a capable color grading tool, and shall be delivered in the highest quality and resolution including, but without limitation to, High Definition (HD) and Ultra High Definition (UHD).
- PSE (Photo-Sensitive Epilepsy) Compliance testing (e.g. Harding, Tektronix, or any other algorithm that complies with ITU-R BT.1702-1 guidelines) must be performed for all animated content.
- **Titles must be delivered as long-play and cannot be split into multiple deliveries**, e.g., if a title is 90 minutes in duration, a single deliverable with a 90-minute duration must be delivered. Two (2) deliverables with 45-minute durations will not be accepted.
- Netflix requires files to be delivered in their **native frame rates**. Netflix defines this to mean what the material was originally shot in or edited with regards to frame rate. If it was originally a film, Netflix requires the native film frame rate (23.976p, 24p or 25p). This is often different from the frame rate that is used for tape, archival storage, and delivery to broadcasters (typically 29.97i df, or 25i). **No 3:2 Pulldown flags allowed.**
- Netflix requires files to be delivered in their **original aspect ratio**. Netflix will not accept a 4x3 version when a 16x9 version was originally created.
- **Partner shall deliver Source Material via Netflix’s [Backlot portal](#), [Content Hub](#) or [VanDAM](#) as directed by your Netflix Representative.** Netflix Backlot is accessible via GUI (graphical user interface) and/or through API (application programming interface) integration with the vendor’s Asset Management System.

3 Primary Digital Asset Specifications

3.1 IMF Packaging

All content must be delivered to Netflix in Interoperable Master Format (IMF). The deliverables must be created according to SMPTE ST 2067-21 IMF Application #2E.

The final IMF package must contain the following:

- 1 Asset Map XML file
- 1 Packing List XML file

- 1 or more Composition Playlist (CPL) XML file(s), representing a single audio language each, containing
 - o 1 image virtual track
 - o 1 surround audio virtual track (6 channels)
 - o 0 or 1 stereo audio virtual track (2 channels)
- Volume Index and Output Profile List XML files may be present, but are not required

Special consideration must be taken for 29.97 or 59.94 IMF deliverables. The total number of frames contained within a 29.97 or 59.94 IMF CPL Segment must be divisible by 5 to preserve integer value audio samples. If necessary, trim the picture and audio by the required number of frames (1, 2, 3, or 4) before wrapping IMF to ensure total frames per Segment are divisible by 5.

Note: Track files containing timed text essence are not currently supported.

3.2 Video

Track files containing image essence mastered in Dolby Vision™ must embed dynamic metadata for color volume transform within a single track file. At a minimum, L1 Dolby Vision™ metadata must be present for every shot.

Note: We do not support material mastered only for HDR-10 distribution.

3.2.1 UHD

UHD Image Track		
Image Frame Width	3840	
Image Frame Height	2160	
Color Encoding	RGB / 4:4:4 / Full Range	
Frame Rates	23.976 / 24 / 25 / 29.97 / 30 / 50 / 59.94 / 60	
Frame Structure	Progressive	
Stereoscopy	Monoscopic	
JPEG 2000 Profile	Frame rates up to 30 fps use: • 4k IMF Single Tile Lossy Profile Mainlevel 6 Sublevel 3 (max 800 Mbit/s) Frame rates above 30 fps use: • 4k IMF Single Tile Lossy Profile Mainlevel 7 Sublevel 4 (max 1600 Mbit/s)	
Dynamic Range	Dolby Vision™ HDR	SDR
Colorimetry	P3 D65 / SMPTE ST 2084 (PQ)	ITU-R BT.709 / D65 / ITU-R BT.1886 (Gamma 2.4)
Pixel Bit Depth	12-bit	10-bit
Mastering Display Color Volume Metadata	SMPTE ST-2086	n/a
Dynamic Metadata for Color Volume Transformation	Dolby Vision™ created by Dolby Mastering Tools SDK v2.6 or greater	n/a

3.2.2 HD

HD Image Track		
Image Frame Width	1920	
Image Frame Height	1080	
Color Encoding	RGB / 4:4:4 / Full Range	
Frame Rates	23.976 / 24 / 25 / 29.97 / 30 / 50 / 59.94 / 60	
Frame Structure	Progressive	
Stereoscopy	Monoscopic	
JPEG 2000 Profile	Frame rates up to 30 fps use: 2k IMF Single Tile Lossy Profile Mainlevel 4 Sublevel 2 (max 400 Mbit/s) Frame rates above 30 fps use: 2k IMF Single Tile Lossy Profile Mainlevel 5 Sublevel 3 (max 800 Mbit/s)	
Dynamic Range	Dolby Vision™ HDR	SDR
Colorimetry	P3 D65 / SMPTE ST 2084 (PQ)	ITU-R BT.709 / D65 / ITU-R BT.1886 (Gamma 2.4)
Pixel Bit Depth	12-bit	10-bit
Mastering Display Color Volume Metadata	SMPTE ST-2086	n/a
Dynamic Metadata for Color Volume Transformation	Dolby Vision™ created with Dolby Mastering Tools SDK v2.6 or greater	n/a

3.3 Audio

3.3.1 Surround Audio

Audio Track (5.1 Surround)	
Sample Rate	48 kHz
Bit Depth	24-bit
Channel Mapping	- Channel 1 – Left - Channel 2 – Right - Channel 3 – Center - Channel 4 – LFE - Channel 5 – Left Surround - Channel 6 – Right Surround

3.3.2 Stereo Audio

Audio Track (Stereo)	
Sample Rate	48 KHz
Bit Depth	24-bit
Channel Mapping	- Channel 1 – Stereo Left - Channel 2 – Stereo Right

4 Audio Specifications

4.1 Audio Prerequisites

Discrete audio is required, except when muxed is inherent to the source, as when audio is delivered via IMF, Quicktime or Atmos BWAV ADM. The only exception is deliveries made to Source Management. See section 5.3 for more details.

Sample Rate and Bit Depth:

48k/24-bit for Original Language Mix or M&E Mix - applies to Stems & Mix Masters.

48k/16-bit or 24-bit for Secondary Language or Audio Description.

Titles with surround sound require 5.1 audio. When 5.1 audio is supplied, preferably an additional 2-channel stereo mix will be supplied.

If conformed 5.1 audio was never created, stereo audio will be accepted. Mono audio is acceptable if the program's original source is mono and no stereo and/or 5.1 mix exists. Mono audio must be duplicated on channels 1 & 2 and delivered as 2-channel.

4.2 Audio Mix Requirements

4.2.1 Original Version from Production and Dubbed Languages

4.2.1.1 Near Field 5.1 Surround Mix - Mastering and Archive

- Use 79dB spl or 82dB spl as your standard reference level for mixing
- Meet a -27 LKFS (+/- 2 LU) dialog-gated using ITU-R BS.1770-1 measured over entire program
- Maintain +18db (-2 dBFS) maximum level (true-peak) over reference of -20 dBFS, achieved by peak limiting and not lowering the mix level
- Provide 5.1 Dialog, Music and Effects stems that equal the 5.1 mix when combined

4.2.1.2 Near Field 2.0 Stereo Mix - Mastering and Archive

- Use 79dB spl or 82dB spl as your standard reference level for mixing
- Provide a separate Lo/Ro or LT/RT mix with -27 LKFS (+/- 2 LU) dialog-gated using ITU-R BS.1770-1 measured over entire program
 - Lo/Ro mix is preferred
- Lo/Ro or LT/RT mix to be mono compatible
- Maintain +18db (-2 dBFS) maximum level (true-peak) over reference of -20 dBFS, achieved by peak limiting and not lowering the mix level
- When down-mixing from 5.1 to create Lo/Ro or LT/RT 2.0 mix:
 - Lower Center channel content by -3db
 - Center channel content should be included in both the LEFT and RIGHT CHANNELS
 - Lower Surround channel content by a minimum of -3db
 - Fold Left and Right Surround content into the corresponding LEFT and RIGHT CHANNELS
 - Optionally include the Low Frequency Effects (LFE) channel between -8db and -12db
 - Fold Low Frequency Effects (LFE) into both the LEFT and RIGHT CHANNELS
 - Low Pass Filter LFE at 200Hz or lower

- All adjustments must be made to the individual 5.1 channels prior to folding down

Note: The resulting 2.0 mix must be checked prior to delivery to ensure it is free of any audio artifacts and if necessary, as determined by Netflix, further adjustments shall be made to prevent them. Any 2.0 delivery must meet the specification prescribed in this section.

4.2.1.3 Near Field Atmos Mix - Mastering and Archive

The Dolby Atmos™ Home Mix must meet the following requirements:

- At a minimum, must be mixed in a 7.1.4 room
- Dolby Certification is not required for Home Atmos mix rooms
- This is a nearfield mix. The most common monitoring levels for nearfield are 79db or 82db
- Must conform to Netflix loudness spec (-27db LKFS +/- 2 LU 1770-1 dialog-gated)
- Peaks should not exceed -2dBFS true-peak. To achieve this, we recommend setting true-peak limiters on all beds and objects at -2.3 or lower. Loudness and peaks should be measured via a 5.1 re-render
- For native Atmos mixes, the 5.1 and 2.0 re-renders must conform to Netflix loudness and Peak specification
- All beds and objects are encoded and will always be audible. A printmaster may utilize a composite of all bed material OR multiple beds, but not both
- For all series/doc/unscripted/comedy content, beds and objects must be separated by Dialog, Music and Effects. For film content, DME separation should be maintained whenever possible
- If an 85 db reference theatrical mix is created, two complete sets of deliverables are required. One for theatrical, one for nearfield
- **Atmos files shall match final IMF picture length exactly** as long-play and not separate reels. Leader and sync pop shall be removed
- FFOA must be left at default - leave "Add FFOA" box unchecked. For pre-3.0 renderer software, leave red dashes in FFOA value
- Final Home Theater Print Master or Localized Mix
 - Dolby Atmos BWAV ADM File
 - Single or multiple LCR, 5.0, 5.1, 7.1 or 7.1.2 bed(s)
 - Tracks 11-128 may be used for Objects or beds
 - 48 kHz, 24-bit

4.2.2 Music & Effects (M&E) Mix

4.2.2.1 Near Field 5.1 Surround M&E Mix

- Provide a separate fully filled 5.1 surround submix containing only Music & Effects (no dialog)
- Ambiences and foley fill should be included to correctly match original mix. Any non-dialog sound present in the main mix must also be represented in the M&E mix
- All levels should mirror what was used for final Mastering and Archive so that any newly added dubbed dialog can be mixed in easily later
- For mixers creating the M&E, Netflix has an M&E Creation Guidelines document available [here](#).

4.2.2.2 Near Field 2.0 Stereo M&E Mix

- Provide a separate fully filled 2.0 stereo submix containing only Music & Effects (no dialog)
- Ambiences and foley fill should be included to correctly match original mix. Any non-dialog sound present in the main mix must also be represented in M&E mix
- Maintain +18db (-2 dBFS) maximum level (true-peak) over reference of -20 dbfs, achieved by peak limiting and not lowering the M&E volume

- When creating Lo/Ro or LT/RT 2.0 mix, follow Near Field 2.0 Stereo Mix spec in section 4.2.1.2
- All levels should mirror what was used for final Mastering and Archive so that any newly added dubbed dialog can be mixed in easily later
- For mixers creating the M&E, Netflix has an M&E Creation Guidelines document available [here](#).

4.2.2.3 Near Field Atmos M&E Mix

The Dolby Atmos™ Home M&E Mix must meet the following requirements:

- At a minimum, must be mixed in a 7.1.4 room
- Dolby Certification is not required for Home Atmos mix rooms
- This is a nearfield mix. The most common monitoring levels for nearfield are 79db or 82db
- Must conform to Netflix loudness spec (-27db LKFS +/- 2 LU 1770-1 dialog-gated)
- Peaks should not exceed -2dBFS true-peak. To achieve this, we recommend setting true-peak limiters on all beds and objects at -2.3 or lower. Loudness and peaks should be measured via a 5.1 re-render
- For native Atmos mixes, the 5.1 and 2.0 re-renders must conform to Netflix loudness and Peak specification
- All beds and objects are encoded and will always be audible. A printmaster may utilize a composite of all bed material OR multiple beds, but not both
- For all series/doc/unscripted/comedy content, beds and objects must be separated by Dialog, Music and Effects. For film content, DME separation should be maintained whenever possible.
- If a 85db reference theatrical mix is created, two complete sets of deliverables are required. One for theatrical, one for nearfield
- **Atmos files shall match final IMF picture length exactly** as long-play and not separate reels. Leader and sync pop shall be removed
- FFOA must be left at default - leave "Add FFOA" box unchecked. For pre-3.0 renderer software, leave red dashes in FFOA value
- M&E Mix
 - Dolby Atmos BWA V ADM File or Pro Tools Recorder Session
 - Single or multiple LCR, 5.0, 5.1, 7.1 or 7.1.2 bed(s)
 - Tracks 11-128 may be used for Objects or beds
 - 48 kHz, 24-bit

4.2.3 Audio Description

4.2.3.1 Near Field 5.1 Surround Mix - Audio Description

- Use 79db spl or 82db spl as your standard reference level for mixing
- Audio Description narration shall be mixed into the center channel
- Meet a -27 LKFS (+/- 2 LU) dialog-gated using ITU-R BS.1770-1 measured over entire program
- Maintain +18db (-2 dBFS) maximum level (true-peak) over reference of -20 dBFS, achieved by peak limiting and not lowering the mix volume
- Provide 5.1 Dialog, Music and Effects stems that equal the 5.1 mix when combined

4.2.3.2 Near Field 2.0 Stereo Mix - Audio Description

- Use 79db spl or 82db spl as your standard reference level for mixing
- Provide a separate Lo/Ro or LT/RT mix with -27 LKFS (+/- 2 LU) dialog-gated using ITU-R BS.1770-1 measured over entire program
 - Lo/Ro mix is preferred
- Lo/Ro or LT/RT mix to be mono compatible

- Maintain +18db (-2 dBFS) maximum level (true-peak) over reference of -20 dBFS, achieved by peak limiting and not lowering the mix level
- When down-mixing to create Lo/Ro or LT/RT 2.0 mix:
 - o Lower Center channel content by -3db
 - o Center channel content should be included in both the LEFT and RIGHT CHANNELS
 - o Lower Surround channel content by a minimum of -3db
 - o Fold Left and Right Surround content into the corresponding LEFT and RIGHT CHANNELS
 - o Optionally include the Low Frequency Effects (LFE) channel between -8 and -12db
 - Fold Low Frequency Effects (LFE) into both the LEFT and RIGHT CHANNELS
 - Low Pass Filter LFE at 200Hz or lower
- All adjustments must be made to the individual 5.1 channels prior to folding down

Note: The resulting 2.0 mix must be checked prior to delivery to ensure it is free of any audio artifacts and if necessary, as determined by Netflix, further adjustments shall be made to prevent them. Any 2.0 delivery must meet the specification prescribed in this section.

5 Secondary Asset Specifications

5.1 Subtitle as a Separate File

Subtitles must encompass spoken dialog intended to be understood that differs in language from the subtitle file. Subtitles must also translate any narrative burned-in text that is in a different language from the subtitle file. In the event of overlapping narrative burned-in text and dialog, precedence should be given to the most plot pertinent message.

For example, for an English original language film, the German subtitle file would contain translation for all the non-German dialog spoken.

- Subtitles must conform, within a half second, to the Primary Video Mezzanine that is delivered or to the Netflix-provided Proxy File. Raster image-based subtitles will not be accepted.
- Subtitle files must not be censored or contain advertisements, placards, overlay branding, or website link callouts.
- All subtitle files need to start at hour 00. Netflix does not accept the subtitle timestamps starting at hour 01 or 10, which is common on tape.
- Include positional data in the subtitle file if the subtitles would otherwise obscure burned-in text.
- All subtitles must contain an individual credit for the author of the text asset. No more than one author credit should be included per asset.
- Subtitler credits must be contained to the last event of the file, must not appear before the end of the main program and must be formatted exactly as they are meant to appear on-screen as required by law in the territory of exhibition.
- TTML format is required for “Right-to-Left” languages such as Arabic and Hebrew.
- Translator credits are NOT required for marketing assets such as trailers, promos, clips, etc.

Framerate Requirements:

- 25 fps media must have subtitles/CC timed to PAL 25 timecode.
- 23.976 (23.98) media must have subtitles/CC timed to either SMPTE 24 timecode or 23.976 Media-Time (timing method must be specified in the delivery metadata as either SMPTE or MEDIA).
- 59.94 fps & 29.97 fps media must have subtitles/CC timed to drop-frame 29.97 timecode.

To ensure the best quality and consistency please reference our [Timed Text Style Guides](#).

Acceptable formats:

- TTML1 (.dfxp, .xml, .ttml) format (see Netflix Timed Text Style Guide for details)
- Lambda cap - for Japanese subtitles only

If your text-based subtitle format is not listed above, please contact your Netflix representative to discuss.

5.1.1 Forced Narrative Subtitles

Netflix requires final AV be delivered without burned-in subtitles. In order to support this non-subtitled source, a list of Forced Narratives, if required for the asset, should be delivered by production using the Netflix-provided template found [here](#).

Examples of Forced Narrative subtitles include:

- Any alternate language dialog and on-screen text
- Inaudible or hard-to-hear dialog
- Dialog or on-screen text that should be called out per creative intent

Example: If an English-language title contains plot-pertinent Japanese content that an English-speaking audience should understand, the final AV should not include burned-in subtitles for those instances. The Forced Narrative template should be filled out with the Japanese dialog translated into English, with preferred punctuation and formatting, as well as corresponding in and out-times for the events.

The Forced Narrative list should include:

- In times and out times for Forced Narrative events
- The event as it should appear on screen, including preferred punctuation

The Forced Narrative list should be uploaded to the Localization bucket in Content Hub as a "Localization Note".

If delivery of a full Forced Narrative subtitle file is required, the technical format must follow the same rules as standard subtitles as denoted in section 5.1.

5.2 Accessibility Files (SDH)

SDH – Subtitles for the Deaf and Hard of Hearing

SDH must encompass the spoken dialog intended to be understood in the content. If the content has multiple spoken languages, they all must be translated and present in the SDH file.

5.2.1 SDH as Separate File

When delivering SDH, it must be submitted in one of the acceptable formats listed below:

- TTML1 (.dfxp, .xml, .ttml) format (see Netflix Timed Text Style Guide for details)
- STL (EBU) - alternate if TTML1 is unavailable
- Lambda cap - for Japanese subtitles only

If your SDH format is not listed above, please contact Netflix to discuss other options.

Additionally, they meet the following:

- Raster image-based SDH will not be accepted.

- SDH files must conform, within a half second, to the Primary Video Mezzanine that is delivered or to the Netflix-provided Proxy File.
- SDH files must not be censored or contain advertisements, placards, overlay branding, website link callouts or company credits unless required by law.
- They must not contain a credit for the SDH creator. They must contain a credit for any translations provided.
- All SDH files need to start at hour 00. Netflix does not accept the SDH timestamps to start at hour 01.
- Include positional data in the SDH file if a caption or subtitle would otherwise obscure burned-in text.
- TTML format is required for “Right-to-Left” languages such as Arabic and Hebrew.

For more detail, please reference our [Timed Text Style Guides](#).

5.3 Alternate Language and Audio Description as a Separate File

If alternate language and/or audio description tracks are required under the Agreement, they shall be conformed to the primary digital audio/video asset delivered. Acceptable formats for alternate-language audio tracks are stereo or multi-channel WAVE Extensible Format (.WAV), or iTunes QuickTime format as described in iTunes Package Asset Specifications. Either format shall be encoded at the highest possible bitrate to maintain fidelity. Alternate language and audio description channel configurations must match those of the original language mix. Deliver both a 5.1 file and a 2.0 file separately if both formats were originally mastered.

Please note: Lectern-style dubbing (single narrator speaking the role of each character) is only acceptable for Polish alternate language audio files and some documentaries or unscripted content. For all other languages and types of content, lip-sync dubbing is required for alternate language audio.

5.3.1 Acceptable Alternate Language and Audio Description Formats (non-QuickTime)

1. **Audio Codec and Container:**
 - a. Uncompressed PCM audio in an interleaved WAVE Extensible Format (.wav)
2. **Bitrate:** Highest bitrate possible (CBR)
3. **Sample rate:** 48 kHz
4. **Bit-depth:** 24-bit or 16-bit
5. **5.1+2.0 Audio Channel Mapping for WAV (.wav)**
 - a. Channel 1 – Left
 - b. Channel 2 – Right
 - c. Channel 3 – Center
 - d. Channel 4 – LFE
 - e. Channel 5 – Left Surround
 - f. Channel 6 – Right Surround
 - g. Channel 7 – Stereo Left
 - h. Channel 8 – Stereo Right
6. **5.1 Audio Channel Mapping for WAV (.wav)**
 - a. Channel 1 – Left
 - b. Channel 2 – Right
 - c. Channel 3 – Center
 - d. Channel 4 – LFE
 - e. Channel 5 – Left Surround
 - f. Channel 6 – Right Surround
7. **2.0 Audio Only Channel Mapping for WAV (.wav)**

- a. Channel 1 – Stereo Left
- b. Channel 2 – Stereo Right

5.3.2 Acceptable Alternate Language and Audio Description Formats (QuickTime)

- iTunes QuickTime format as described in iTunes Package Asset Specifications is acceptable.
- Please pay special attention to the audio track / channel assignments defined in the iTunes Package Asset Specifications.
- All channels must be tagged in accordance with the iTunes spec.
- Also, note that the labels “Left Total” and “Right Total” are used to label the stereo mix channels only used when 5.1 + 2.0 audio is included in a single file (Options 1-4). If stereo only files are delivered (Options 5 & 6) then the labels “Left” and “Right” are used since there is no ambiguity.

1. Audio Codec

- a. Uncompressed LPCM – 16-bit or 24-bit, 48 kHz (Big or Little Endian)

2. 5.1 Surround + 2.0 Stereo Audio Channel Mapping for QuickTime .mov

- a. **Option 1 (3 tracks)**
 - i. **Track 1 (6 channels)**
 - 1. Channel 1 – Center
 - 2. Channel 2 – Left
 - 3. Channel 3 – Right
 - 4. Channel 4 – Left Surround
 - 5. Channel 5 – Right Surround
 - 6. Channel 6 – LFE
 - ii. **Track 2 (1 channel)**
 - 1. Channel 1 – Stereo Left
 - iii. **Track 3 (1 channel)**
 - 1. Channel 1 – Stereo Right
- b. **Option 2 (8 tracks)**
 - i. **Track 1 (1 channel)**
 - 1. Channel 1 – Left
 - ii. **Track 2 (1 channel)**
 - 1. Channel 1 – Right
 - iii. **Track 3 (1 channel)**
 - 1. Channel 1 – Center
 - iv. **Track 4 (1 channel)**
 - 1. Channel 1 – LFE
 - v. **Track 5 (1 channel)**
 - 1. Channel 1 – Left Surround
 - vi. **Track 6 (1 channel)**
 - 1. Channel 1 – Right Surround
 - vii. **Track 7 (1 channel)**
 - 1. Channel 1 – Stereo Left
 - viii. **Track 8 (1 channel)**
 - 1. Channel 1 – Stereo Right
- c. **Option 3 (2 tracks)**
 - i. **Track 1 (6 channel)**
 - 1. Channel 1 – Left
 - 2. Channel 2 – Right
 - 3. Channel 3 – Center
 - 4. Channel 4 – LFE

- 5. Channel 5 – Left Surround
 - 6. Channel 6 – Right Surround
 - ii. **Track 2 (2 channel)**
 - 1. Channel 1 – Stereo Left
 - 2. Channel 2 – Stereo Right
 - d. **Option 4 (7 tracks)**
 - i. **Track 1 (1 channel)**
 - 1. Channel 1 – Left
 - ii. **Track 2 (1 channel)**
 - 1. Channel 1 – Right
 - iii. **Track 3 (1 channel)**
 - 1. Channel 1 – Center
 - iv. **Track 4 (1 channel)**
 - 1. Channel 1 – LFE
 - v. **Track 5 (1 channel)**
 - 1. Channel 1 – Left Surround
 - vi. **Track 6 (1 channel)**
 - 1. Channel 1 – Right Surround
 - vii. **Track 7 (2 channel)**
 - 1. Channel 1 – Stereo Left
 - 2. Channel 2 – Stereo Right
3. **2.0 Stereo Audio Only Channel Mapping for QuickTime .mov**
- a. **Option 1 (2 tracks)**
 - i. **Track 1 (1 channel)**
 - 1. Channel 1 – Stereo Left
 - ii. **Track 2 (1 channel)**
 - 1. Channel 2 – Stereo Right
 - b. **Option 2 (1 track)**
 - i. **Track 1 (2 channel)**
 - 1. Channel 1 – Stereo Left
 - 2. Channel 2 – Stereo Right

5.3.3 Dub Cards / Dub Credits

If dubbed audio files are sent, Partner shall deliver such Dub Cards to Netflix formatted exactly as they are to appear on-screen. No extraneous information can be included beyond studio name, cast and other key personnel or as otherwise required by law in the territory of exhibition.

Acceptable Dub Card Formats

- **Image Files**
 - o Each Image must represent one frame of dubbed actor credits
 - o Only one language can be represented on a dub card and the dubbed language must be specified in the dub card text
 - Example: "Version Française"
 - o We accept the following flat file types in order of preference (no layers or pages):
 - TIFF
 - PNG
 - BMP
 - JPG
 - o Minimum Resolution: 640x480
 - o Pixel Aspect Ratio (PAR): 1:1 square pixels

In cases where multiple dub cards are needed for an individual language, the multiple dub cards must be stored in a ZIP archive.

- **ZIP Archive**
 - o Each Image must meet the image file spec described above
 - o Each image must match the file format and resolution of the other images in the ZIP archive
 - o All images in ZIP archive must be for the same language
 - o Images will be rendered to screen in alpha-numeric sorted order (case insensitive)
 - o Acceptable Compression modes: “DEFLATE” or “STORE”
 - Note: WinZip and WinRAR use DEFLATE by default
 - o **Examples:** (note that the ZIP file name does not need to adhere to any specific convention. Image files contained in the ZIP archive must be named in case-insensitive alphanumeric order)
 - Brazilian Portuguese
 - ZIP File name: SomeTitle_pt-BR.zip
 - Image names: dubCard01.tiff, dubCard02.tiff, dubCard03.tiff
 - French Parisian
 - ZIP File name: FrenchDubCards.zip
 - Image names: FrenchActors01.png, FrenchActors02.png, FrenchActors03.png

6 Movie Content Metadata

Movie metadata is required for all content and is to be submitted via the approved Netflix Metadata Excel Template to your Netflix Representative.

Partner shall provide all film/television ratings information that exists for each Title with respect to the licensed Territory and the United States, if any.

All dubbed and/or subtitled content is to be accompanied with the relevant metadata in the original and alternate language to support the source delivered.

7 Delivery Platforms

Partner shall deliver Source Material via [Backlot](#), [Content Hub](#) or [VanDAM](#) as determined by your Netflix Representative. Backlot is accessible via GUI (graphical user interface) and/or through API (Application Programming Interface) integration with the vendor’s Asset Management System.

Netflix uses Aspera exclusively for network delivery. Partner must push content to Netflix via Aspera. Netflix will not pull content. Please discuss this with your Netflix Representative if you have any questions about this requirement.

7.1 Backlot Delivery UI

Netflix supports delivery of source assets via the Backlot Delivery UI which utilizes the Aspera Connect Plugin. Information on how to use this feature can be located on the Partner Help Center [here](#).

7.2 Backlot Requests & Delivery Management API

Delivery via the Backlot Request & Delivery Management API is available. If you are interested in delivering via this method please review the documentation located [here](#).

8 Marketing/Publicity Assets and Source Materials

Static Images and AV provided will be used within the Netflix service (including, without limitation, on all Netflix Ready Devices, through the website, User Interfaces, applications, blogs, social networking sites and our communications with members and potential members) and in marketing materials promoting the titles' availability on Netflix. Netflix shall have the right, but not the obligation, to modify (e.g., edit size, placement, crop, or make other alterations to) and/or create any additional artwork discussed in this Section 8.

In addition, all artwork layered photoshop document assets (hereinafter called "Fully Layered PSD") shall be provided in a completely layered format that allows background images, foreground images, every cast member to be re-positioned or otherwise modified.

All art should be provided in localized format (i.e., as it is marketed in the Territory) including Title Treatments.

For any specific questions or concerns related to any asset requirements listed below, please reach out to your Netflix Representative.

8.1 Key Art – Vertical, Horizontal

Asset	Specifications	File Format	Requirements
Key Art Vertical	7200 (w) x 10088 (h) 300 ppi RGB (8-bits/channel)	Fully Layered PSD	Vertical artwork shall not have title treatment in the lower 1/5 of the image area.
Key Art Horizontal	9000 (w) x 5063 (h) 300 ppi RGB (8-bits/channel)	Fully Layered PSD	Horizontal artwork shall not have the title treatment in the lower 1/5 of the image area.

8.2 Title Treatment and Title Card

Each Title Treatment asset should adhere to the following guidelines:

- Dark and Light versions on transparent background
- Provided font names and related files
 - If custom created font, Licensor to supply alphabet reference file.
- No extraneous text such as quotes, taglines, credits, copyrights, billing blocks.
- No extraneous logos such as DVD, Blu-ray, award laurels, studio logos.
- No weapons pointing at camera, extreme gore, or nudity.
- Fully Layered PSD

Asset	Specifications	File Format	Requirements
Title Treatment Unstacked (one-line)	5000 (w) x 1125 (h) 300 ppi RGB (8-bits/channel)	Vector EPS, AI, or layered PSD	Delivered in Dark and Light versions with Font Names and alphabet reference files; Artwork shall not include any borders, frames, logos, copyright lines, taglines, or quotes.

Title Treatment Stacked (two-line)	5000 (w) x 2000 (h) 300 ppi RGB (8-bits/channel)	Vector EPS, AI, or layered PSD	Delivered in Dark and Light versions with Font Names and alphabet reference files; Artwork shall not include any borders, frames, logos, copyright lines, taglines, or quotes.
Title Card	3050 (w) x 1715 (h) 300 ppi RGB (8-bits/channel)	Fully Layered PSD	Title Cards shall feature a legible, prominent title treatment on a contrasting background devoid of characters or objects.

8.3 Stills or Layered Stills

Asset	Specifications	File Format	Requirements
Stills - Live Action	3840 (w) x 2160 (h) 300 ppi RGB (8-bits/channel)	High-Resolution JPG or TIFFs	No text, logos, or copyright lines. No weapons pointing at camera, extreme gore, or nudity.
Layered Stills - Animated	Largest Derivative 5000 (w) x 2813 (h) 300 ppi, RGB (8-bits/channel)	Fully Layered, High-Resolution PSD	No text, logos, or copyright lines. No weapons pointing at camera, extreme gore, or nudity. Examples found here .

8.4 Character Poses

Asset	Specifications	File Format	Requirements
Character Poses - Animated	Largest Derivative 5000 (h) x 5000 (w) 300 ppi, RGB (8-bits/channel)	Hi-res layered PSD, full body, alpha channel or transparent TIFF	Artwork shall not be obstructed by text, objects or other characters. Artwork shall not include any borders or frames. Examples found here .

8.5 Unit Photography

Asset	Specifications	File Format	Requirements
Unit Photos	Largest Derivative 5000 (h) x 5000 (w) 300 ppi RGB (8-bits/channel)	Original RAW files & corresponding high resolution JPGs or TIFF	Photographer credit, color transparencies, color negatives and contact sheets are to be included with delivery. Examples found here .

8.6 Gallery/Cast Photography

Asset	Specifications	File Format	Requirements
Gallery Photography	Largest Derivative 5000 (h) x 5000 300 ppi RGB (8-bits/channel)	Original RAW files & corresponding high resolution JPGs or TIFF	Photographer credit, color transparencies, color negatives and contact sheets are to be included with delivery. Examples found here .

8.7 Backgrounds & Environmentals

Asset	Specifications	File Format	Requirements
Environmentals & Backgrounds	Largest Derivative 5000 (h) x 5000 300 ppi RGB (8-bits/channel)	Layered PSD or High-Resolution JPG	Plates or background images shall be devoid of any characters. No title treatment, text, logos, or copyright lines. Examples found here .

8.8 AV Trailers & Promos

For each AV asset, Licensor/Producer must deliver:

- Texted and Textless versions
- Music Cue Sheet
- Dialog List (or a Spotting List / EDL)
 - Preferably a **dialog list** that includes the name of the character(s) speaking, dialog, and the timecode (TCs) referencing where from the full content (incl. episode numbers, if applicable) the dialog was pulled from
- AE Collect/Graphics Package for Graphic Cards
- Full Audio Splits and DME stems
- Audio Printmaster
 - Need a separate delivery of the Printmaster in the same mix of the original film or episodes:
 - 5.1 (6-channel), 2.0 (2-channel), 5.1 + 2.0 (8-channel)

If requested, Licensor/Producer shall localize (i.e. subtitle, dub) the promotional videos. If more promotional videos (including, without limitation, videos intended for social media) beyond the minimum are created, Licensor/Producer shall promptly deliver such videos.

Asset	Video Specifications	File Format	Requirements
Trailer	In highest resolution available/native to show	Preferred: SMPTE ST 2067-21 (Interoperable Master Format - Application #2E) Alternative: ProRes 422 HQ	Trailer video frame rate and codec should match the original film or episodes, provided in Textless and Texted versions, with music cleared for usage worldwide in perpetuity, and full Audio Splits and DME stems.

AV Promos	In highest resolution available/native to show	Preferred: SMPTE ST 2067-21 (Interoperable Master Format - Application #2E) Alternative: ProRes 422 HQ	Trailer video frame rate and codec should match the original film or episodes, provided in Textless and Texted versions, with music cleared for usage worldwide in perpetuity, and full Audio Splits and DME stems.
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8.9 EPK Source Materials

Asset	Video Specifications	File Format	Requirements
EPK Source Materials	In highest resolution available/native to show	ProRes 422 Quicktime	Original, unedited EPK footage and source materials as well as edited by the Partner

8.10 Custom Animation or Video + Voiceover

Asset	Specifications	File Format	Requirements
Custom Animation	In highest resolution available/native to show	ProRes 422 Quicktime	Audio Splits or 5.1 Tracks
Custom Video	In highest resolution available/native to show	ProRes 422 Quicktime	Audio Splits or 5.1 Tracks (Note: Raw 4K Footage + Audio Splits preferred)
Custom Voiceover	N/A	WAV and/or MP3	Minimum of 2 minutes in length

9 Appendix

9.1 Language Codes

If the desired language code is not found in the table below please contact your Netflix Representative.

Language	Code	Language	Code
Acoli	ach	Korean	ko
Afrikaans	af	Konkani	kok
Amharic	am	Kashmiri	ks
Arabic	ar	Kurdish	ku
Arabic (United Arab Emirates)	ar-AE	Kyrgyz	ky
Arabic (Bahrain)	ar-BH	Latin	la

Arabic (Djibouti)	ar-DJ	Lango (Uganda)	laj
Arabic (Algeria)	ar-DZ	Lingala	ln
Arabic (Egypt)	ar-EG	Lithuanian	lt
Arabic (Western Sahara)	ar-EH	Latvian	lv
Arabic (Eritrea)	ar-ER	Maithili	mai
Arabic (Israel)	ar-IL	Mandingo	man
Arabic (Iraq)	ar-IQ	Masai	mas
Arabic (Jordan)	ar-JO	Mende	men
Arabic (Comoros)	ar-KM	Malagasy	mg
Arabic (Kuwait)	ar-KW	Maori	mi
Arabic (Lebanon)	ar-LB	Macedonian	mk
Arabic (Libya)	ar-LY	Malayalam	ml
Arabic (Morocco)	ar-MA	Mongolian	mn
Arabic (Mauritania)	ar-MR	Manipuri	mni
Arabic (Oman)	ar-OM	Marathi	mr
Arabic (Palestine)	ar-PS	Malay	ms
Arabic (Qatar)	ar-QA	Maltese	mt
Arabic (Saudi Arabia)	ar-SA	Multiple languages	mul
Arabic (Sudan)	ar-SD	Burmese	my
Arabic (Somalia)	ar-SO	Mayan	myn
Arabic (South Sudan)	ar-SS	Hokkien	nan
Arabic (Syria)	ar-SY	Taiwanese	nan-TW
Arabic (Chad)	ar-TD	Norwegian	nb
Arabic (Tunisia)	ar-TN	North Ndebele	nd
Arabic (Yemen)	ar-YE	Nepali	ne
Aramaic	arc	Newari	new
Assamese	as	Dutch	nl
Bashkir	ba	Flemish	nl-BE
Berber	ber	South Ndebele	nr
Bulgarian	bg	Northern Sotho	nso
Bambara	bm	Nyanja	ny
Bangla	bn	Oromo	om
Bantu	bnt	Odia	or
Tibetan	bo	Punjabi	pa
Bodo	brx	Polish	pl
Bosnian	bs	Pashto	ps
Catalan	ca	Portuguese (Portugal)	pt

Chechen	ce	Portuguese (Brazil)	pt-BR
Cebuano	ceb	Romansh	rm
Cheyenne	chy	Romanian	ro
Czech	cs	Romany	rom
Welsh	cy	Russian	ru
Danish	da	Kinyarwanda	rw
German	de	Sanskrit	sa
Dinka	din	Santali	sat
Dogri	doi	Sindhi	sd
Dzongkha	dz	Sherdukpen	sdp
Efik	efi	Sinhala	si
Greek	el	Slovak	sk
Elamite	elx	Slovenian	sl
English	en	Samoan	sm
English (Australia)	en-AU	Sami	smi
English (Britain)	en-GB	Shona	sn
Esperanto	eo	Somali	so
Spanish (Latam)	es	Songhai	son
Spanish (Spain)	es-ES	Albanian	sq
Estonian	et	Serbian	sr
Basque	eu	Serbian (Cyrillic)	sr-Cyrl
Persian	fa	Serbian (Latin)	sr-Latn
Dari	fa-AF	Swati	ss
Fulah	ff	Southern Sotho	st
Finnish	fi	Swedish	sv
Filipino	fil	Swahili	sw
Faroese	fo	Tamil	ta
French	fr	Telugu	te
French (Canada)	fr-CA	Thai	th
Irish	ga	Klingon	tlh
Scottish Gaelic	gd	Tamashek	tmh
Galician	gl	Tswana	tn
Swiss German	gsw	Tongan	to
Gujarati	gu	Turkish	tr
Hausa	ha	Taroko	trv
Hawaiian	haw	Tsonga	ts
Hebrew	he	Tuvian	tyv

Hindi	hi	Ugaritic	uga
Croatian	hr	Ukrainian	uk
Hungarian	hu	Urdu	ur
Armenian	hy	Venda	ve
Indonesian	id	Vietnamese	vi
Igbo	ig	Wolof	wo
Icelandic	is	Bulbovian	xa
Italian	it	Xhosa	xh
Inuktitut	iu	Yiddish	yi
Japanese	ja	Yoruba	yo
Jamaican Patois	jam	Cantonese	yue
Javanese	jv	Zapotec	zap
Georgian	ka	Mandarin	zh
Khasi	kha	Simplified Chinese	zh-Hans
Kikuyu	ki	Traditional Chinese	zh-Hant
Kazakh	kk	Mandarin (Taiwan)	zh-TW
Khmer	km	Zulu	zu
Kannada	kn	No linguistic content	zxx

9.2 Change Log History

Change Log OC-3-3 (2019-07-08)

- Updated Photosensitive Epilepsy Detection to adhere to ITU-R BT.1702-1 instead of Ofcom
- Corrected HD J2K profile requirements
- Section 5.3.1 - removed LCR film mapping as acceptable
- Section 5.3.1 - removed LoRo and replaced with Stereo Left and Stereo Right
- Section 4.2.1.2 - added a range for the LFE track & that any 2.0 delivery must meet the spec
- Section 4.2.3.2 - added a range for the LFE track & that any 2.0 delivery must meet the spec
- Removed support for AES3 LPCM in MPEG-2 Transport Stream (.mpg)
- Updated accepted Language Codes
- Minor editorial changes

Change Log OC-3-2-1 (2019-06-04)

- Corrected J2K profile requirements

Change Log OC-3-2 (2019-06-03)

- Clarified that alternate language deliveries must be delivered muxed
- Clarified language regarding long-play file requirement
- Removed Timecode Start requirement for MXF files
- Removed requirement for OPL and VOLINDEX to be present in IMF package
- Removed requirement to reference all files from CPL
- Removed support for YCbCr color/legal range encoding
- Removed support for 4K/2K (full container) resolution
- Added requirement for IMF packages to be generated from VDM or directly out of a capable color grading tool

- Added requirement for surround sound tracks to be present in all CPLs
- Added support for multiple CPLs to be present in an IMF Package
- Added Atmos Home Theater Printmaster, Localized Mix and M&E specifications
- Added all asset type requests for Marketing, Publicity, and Creative Production
- Added Atmos Specifications for Near Field M&E
- Added Atmos Specifications for Near Field Mix
- Updated accepted Language Codes
- Removed Tagalog and Norwegian (Nynorsk) from the accepted Language Codes
- Removed File Naming Conventions
- Added Content Hub and VanDAM as delivery platforms
- Added further context to the Objective
- Removed Aspera whitelist guidelines
- Minor editorial changes

Change Log OC-3-1 (2018-7-26)

- Added additional 2.0 creation specifications - Center must be dropped 3db, surrounds minimum 3db
- Added special considerations for IMF frame rates, page 5
- Added requirement for alternate language and audio description channels to match original language channel configuration, section 5.3
- Removed language requiring that 2.0 mixes not be fold downs, in section 4.1. Added sample rate and bit depth information
- Section 4.2.3.2, updated 2.0 creation guidelines

Change Log OC-3-0 (2017-6-7)

- Updated “non-subtitled” from a preference to a requirement
- Added PSE Compliance testing (Harding) for all animated content
- Added Output Profile List XML File as a required asset for IMF packages
- Updated minimum Dolby Vision Mastering SDK version to 2.6
- Removed ProRes 422 HQ as an acceptable primary AV deliverable for HD
- Updated Artwork section with new requirements
- Added Lo/Ro as acceptable stereo format
- Updated Content Partner Portal section
- Changed overall title of document
- Updated IMF requirements to support 2K/HD resolutions
- Updated IMF requirements to use IMF Mainlevel. BCP Levels have been replaced in SMPTE IMF standards by IMF Mainlevels
- IMF Packages no longer need to be ZIP’ed
- Updated ProRes 422 (HQ) specs to support additional 2K resolutions
- Updated ProRes 422 (HQ) specs to restrict to Rec.709 color space
- Removed XML Manifest requirements since XML Manifests are no longer needed nor supported by Backlot
- Updated the Note in Alternate Language Audio Files regarding lectern dubbing
- Removed EBU-STL files as an acceptable
- Updated Artwork section with new requirements
- Minor editorial changes

Change Log OC-2-5 (2015-7-31)

- Updated audio mix specifications with LKFS units rather than dialnorm since BT.1770-3 is used
- Specified LKFS measurement should be across entire program
- Added +/- 2 threshold variance for LKFS measurement
- Specified the use of true peak for peak values

- Updated IMF requirements to support creation of Dolby Vision IMF Packages

Change Log OC-2-4 (2014-10-09)

- Update 4K / UHD IMF spec by requiring luminance to be in HEAD/LEGAL range and removed full range as a requirement.

Change Log OC-2-3 (2014-10-01)

- Reorganized document to be clearer
- Significant update to Artwork section

Change Log OC-2-2 (2014-8-05)

- Removed 2 erroneous example file names

Change Log OC-2-1 (2014-8-05)

- Removed AAC (.mp4) as acceptable audio format
- Added specifications for Audio Description files

Change Log OC-2-0 (2014-7-03)

- Deprecated 4K DPX and replaced with IMF specs
- Added detail for audio mix specifications to use ITU-R BS.1770-3 for reference
- Updated to new Netflix logo

Change Log OC-1-9 (2014-4-29)

- Updated Dub Credits to read Dub Cards / Dub Credits
- Updated Dub Cards / Dub Credits deliverable requirements to image format

Change Log OC-1-8 (2014-4-14)

- Updated Dub Cards to read Dub Credits

Change Log OC-1-7 (2014-4-2)

- Added Dub Card specifications
- Updated description to Netflix Timed-Text Style Guide for quality and consistency purposes

Change Log OC-1-6 (2014-3-21)

- Fixed issue with heading numbering for Primary Digital Asset Specifications section
- Updated file names to support 23976

Change Log OC-1-5 (2014-3-1)

- Integrated 4K spec

Change Log OC-1-4 (2014-2-25)

- Added Audio Mix Requirements
- Changed wording of Primary Mezzanine Audio Prerequisites (no "simple" mixdowns allowed)
- Reformatted layout

Change Log OC-1-3 (2014-2-6)

- Removed SMPTE24 conformance since all timed-text should be synced to video playback (media time aka wall clock)
- Added acceptance of WAV file for secondary audio
- Updated Artwork section

Change Log OC-1-2 (2014-1-17)

- Added Movie Content Metadata section
- Added Artwork section

Change Log OC-1-1 (2013-11-22)

- Removed references to CC in lieu of SDH
- Cleaned up Language Code layout to two columns

Change Log OC-1-0 (2013-10-23)

- First version